

Uniformly Elliptic ABP

1. ENTROPIES

Assume that $v > 0$ and b are functions such that v and bv are both caloric. Then

$$(1) \quad (\partial_t - \Delta)(vb^2) = -2v|\nabla b|^2.$$

We also record

$$(2) \quad (\partial_t - \Delta)v \log(1 + v) \geq -\frac{|\nabla v|^2}{1 + v}.$$

These seem to be instances of a general entropy formula due to Michel-Mischler-Perthame, cited in Lukas' note.

2. ENERGY ESTIMATE FOR ADJOINT

Assume that $A(x) \in \text{Sym}_{n \times n}$ has eigenvalues in $[\lambda, \lambda^{-1}]$ and that $v > 0$ solves

$$(3) \quad \text{div div}(A(x)v) = -f, \quad f \geq 0.$$

Then

$$(4) \quad \int \frac{|\nabla v|^2}{1 + v} \leq C(\lambda) \int v |\text{div} A|^2 + C(\lambda) \int f \log(1 + v).$$

Proof: Test (3) with $\log(1 + v)$, integrate by parts, use Cauchy-Schwarz.

3. ABP

Take A, v, f as in the previous section and assume that v and f both have mass

1. The key is to recognize the left side of (4) as the dissipation rate of entropy, and the right as the dissipation rate of a bounded quantity.

Let the subscript h denote evolution by heat flow for time h , and let

$$B^h := \frac{(Av)_h}{v_h} \in [\lambda I, \lambda^{-1} I].$$

Then

$$\text{div div}(B^h v_h) = -f_h.$$

By (4) and the fact that f has unit mass we have

$$\int \frac{|\nabla v_h|^2}{1 + v_h} \leq C(n, \lambda) \int v_h |\nabla B^h|^2 + C(\lambda) \log(1 + \|v_h\|_{L^\infty}).$$

Integrating in h from 0 to 1, applying (1) and (2), and using that $|v_h| \leq C(n)h^{-n/2}$ gives

$$\int v \log(1 + v) < C(n, \lambda).$$